



## Regular article

## Infrared dim moving target tracking via sparsity-based discriminative classifier and convolutional network

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## HIGHLIGHTS

- An representation model via SDC, CNGM and an efficient prior model is presented.
- Collaborative confidence values are utilized to generate particles' weights.
- The update scheme not only accounts for appearance variations but also alleviates the migration.
- This algorithm yields better performance than several state of the art algorithms.

## ARTICLE INFO

## Article history:

Received 5 September 2016

Revised 24 July 2017

Accepted 5 September 2017

Available online 6 September 2017

## Keywords:

Infrared image

Dim target tracking

Sparse representation

Convolutional network

Particle filter

## ABSTRACT

Infrared dim and small target tracking is a great challenging task. The main challenge for target tracking is to account for appearance change of an object, which submerges in the cluttered background. An efficient appearance model that exploits both the global template and local representation over infrared image sequences is constructed for dim moving target tracking. A Sparsity-based Discriminative Classifier (SDC) and a Convolutional Network-based Generative Model (CNGM) are combined with a prior model. In the SDC model, a sparse representation-based algorithm is adopted to calculate the confidence value that assigns more weights to target templates than negative background templates. In the CNGM model, simple cell feature maps are obtained by calculating the convolution between target templates and fixed filters, which are extracted from the target region at the first frame. These maps measure similarities between each filter and local intensity patterns across the target template, therefore encoding its local structural information. Then, all the maps form a representation, preserving the inner geometric layout of a candidate template. Furthermore, the fixed target template set is processed via an efficient prior model. The same operation is applied to candidate templates in the CNGM model. The online update scheme not only accounts for appearance variations but also alleviates the migration problem. At last, collaborative confidence values of particles are utilized to generate particles' importance weights. Experiments on various infrared sequences have validated the tracking capability of the presented algorithm. Experimental results show that this algorithm runs in real-time and provides a higher accuracy than state of the art algorithms.

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## 1. Introduction

Infrared dim and small target tracking is an important issue in image processing and has been widely used in military and civilian [1,2]. The purpose of infrared target tracking is similar to visual target tracking, which aims to establish a target's position and to track it in subsequent frames. In general, the motion state of a dim moving target can be random and uncertain. Besides, a dim moving target in sensed videos appears as a point with little shape and

textural information, due to the fact that it is far away from the optical system. Moreover, infrared dim moving targets with low Signal-to-Clutter Ratio (SCR) usually submerge in complicated backgrounds, which is the crucial challenge [3,4]. Therefore, the accurate detection and tracking of a dim target is considered as a difficult work.

Many researchers have made their contributions to infrared dim target detection and tracking [5–12], such as Template Matching Tracking (TMT) algorithm [6,7], optical flow algorithm [8], Mean Shift (MS) algorithm [9,10], Moving Pipeline Filtering (MPF) algorithm [11], and Particle Filter (PF) algorithm [12]. To improve the performance of the TMT algorithms, Liu et al. pro-

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posed a tracking framework based on the template matching combined with the Kalman prediction [13]. Such framework uses the projection coefficients of PCA as templates and measures the matching degree by using nonlinear correlation coefficients, which constitute the detection phase and provide more precise measurement for the Kalman prediction in the next tracking step. However, severe background clutter may lead to the cumulative error. The MS algorithm [14,15] represents infrared target appearances by kernel weighted gray histogram and utilizes the mean shift procedure to identify the most likely position of the target in the next frame. Yet kinds of complex backgrounds make the target feature less discriminative, resulting in an inaccurate parameter estimation in the tracking model. In PF-based algorithms, target tracking is considered as a recursive Bayesian estimate problem, and estimated states are represented by some parameters of tracked targets, such as position and velocity [16,17]. To cope with the uncertainty of a dim target, Huang et al. proposed an efficient algorithm for tracking a small object [17], it is initialized using a strong detector, which is created from the shape analysis of foreground blobs, and the ambiguity of the template matching can be handled with a PF-based framework. Nevertheless, this algorithm does not focus on giving a sufficient account of a target's feature once the target submerges in the cluttered background.

To efficiently extract the feature for the appearance model, Zhang et al. [18] proposed a Fast Compressive Tracking (FCT) algorithm, which constructed a very sparse measurement matrix. Then, the tracking task is formulated as a binary classification via a naive Bayes classifier. Besides, Li et al. combined the appearance model in the FCT algorithm with the popular  $\ell_1$  tracker [2]. Besides, over-complete dictionary based sparse representation is an effective image representation method that can capture geometrical features of infrared small targets by the redundancy of the dictionary. Therefore, Wan et al. [19] proposed a sparse representation model with a non-negative constraint to detect the dim target. For less computation, a preprocessing method based on frequency saliency detection is adopted to extract the suspected regions containing the target. Moreover, Li et al. presented a PF-based tracking algorithm [20]. The discriminative over-complete dictionary is utilized to enlarge the difference between the target particle and the background particle. Further, the particle discriminative sparse representation improves the accuracy of target motion estimation by means of heightening the weight of target particles. The shortcoming is that particles may degrade into a single value.

This paper presents an infrared dim moving target tracking algorithm via a collaborative representation, which integrates a Sparsity-based Discriminative Classifier (SDC) [21] and a Convolutional Network-based Generative Model (CNGM) [22]. A typical tracking system consists of three components [23], i.e., an appearance model, a motion model which needs to be updated, and a search strategy for finding the most likely position in the current frame. To represent the target, the holistic image intensity and the local structural information are extracted from the object region. To restrain the negative influence of the background clutter, target templates and candidate templates in the CNGM model are processed via a prior model. As an appearance representation, the discriminative SDC model takes the background information into consideration, and the generative CNGM model exploits the inner geometric layout information of a target template. As a motion model, the particle filter framework is adopted to predict the target position. At last, the collaborative weight of each candidate template is obtained, and templates with larger values are selected to estimate the target position in the current frame. Experimental results validate that the presented algorithm can significantly improve the accuracy over infrared image sequences, in comparison with several state of the art algorithms.

The remainder is organized as follows. In Section 2, the related work referring to particle filter is reviewed, and the motivation of our work is described. In Section 3, a prior model, which can enhance the target template, is applied to the sparsity-based discriminative classifier model. Then, the Convolutional Network-based Generative Model is described. Besides, candidate templates are all processed with the prior to restrain the negative influence of the background clutter. Then, the SDC-CNGM algorithm is presented to deal with the infrared dim moving target tracking. In Section 4, qualitative and quantitative results are introduced to verify the performance of the presented algorithm over numerous infrared image sequences. Finally, conclusions are drawn in Section 5.

## 2. Related works

### 2.1. The particle filter framework

The particle filter [16,19] is a Bayesian sequential importance sampling technique and provides a rigorous general framework for dynamic state estimation problems. From Bayesian perspective, the problem associated with tracking is recursively calculating some degree of belief in state  $x_k$  and constructing the posterior Probability Density Function (PDF)  $p(x_k|z_{1:k})$  of the state, with  $z_{1:k} = \{z_1, \dots, z_k\}$  denoting the cumulative set of measurements up to frame  $k$ .

The key step of this framework is prediction and update. Suppose the required PDF  $p(x_{k-1}|z_{1:k-1})$  at frame  $(k-1)$  is available, the prior probability density  $p(x_k|z_{1:k-1})$  at frame  $k$  is predicted by

$$p(x_k|z_{1:k-1}) = \int p(x_k|x_{k-1})p(x_{k-1}|z_{1:k-1})dx_{k-1} \quad (1)$$

where  $p(x_k|x_{k-1})$  represents the state transition probability density to predict the prior state at frame  $k$ .

Once the observation  $z_k$  at frame  $k$  is obtained, the posterior PDF  $p(x_k|z_{1:k})$  of the dim target state can be updated using Bayes' rule

$$p(x_k|z_{1:k}) = p(z_k|x_k)p(x_k|z_{1:k-1})/p(z_k|z_{1:k-1}) \quad (2)$$

where  $p(z_k|x_k)$  is described by an observation equation and  $p(x_k|z_{1:k-1})$  denotes a normalizing denominator.

At frame  $k-1$ , a group of random samples  $\{x_{k-1}(i)\}_{i=1}^N$  with importance weights  $w_{k-1}^i$  are obtained from the PDF  $p(x_{k-1}|z_{1:k-1})$ . Then, the PF algorithm propagates and updates these samples to obtain a group of new samples  $\{x_k(i)\}_{i=1}^N$ , which are approximately distributed as  $p(x_k|z_{1:k})$ .

In the prediction stage, a particle set  $x_k^*(i)$  at time  $k$  is obtained with each sample  $x_{k-1}(i)$ . On receipt of the measurement  $z_k$ , the likelihood of each prior sample is evaluated and a normalized weight  $q_i$  for each predicated sample is expressed by

$$q_i = p(z_k|x_k^*(i))/\sum_{j=1}^N p(z_k|x_k^*(j)) \quad (3)$$

Therefore, prior particles can be re-sampled to generate an unweighted particle set according to importance weights, helping to avoid degeneracy.

### 2.2. Motivation of our work

The representation schemes for object tracking mainly consist of holistic templates and local structural information. While most tracking algorithms use either holistic or local representations, the presented approach exploits the collaborative strength of both schemes. Most tracking methods use rectangle to represent the tracking result, yet the pixels within the tracking rectangle are not all from foreground. As a result, the local representation-based classifier may be affected when updated with the back-

ground patches as positive samples. On the contrary, the holistic templates are often distinct to be foreground or background. Thus, the holistic templates are more suitable for discriminative models. Meanwhile, local representations are more amenable for generative models because of their flexibility. Therefore, a collaborative model that integrates a discriminative classifier based on holistic templates and a generative model using local representations is presented for dim-small target tracking.

### 3. The proposed SDC-CNGM tracking

In this section, a target tracking algorithm is proposed in details. Specially, the discriminative model and generative model are described. The prior model is introduced to enhance the target template. Besides, the update method for the appearance representation is then presented.

#### 3.1. The improved SDC model

The SDC model is developed based on the assumption that a target template can be represented by the linear combination of a positive template set  $A_+$ , while a background template can be represented by the combination of a negative template set  $A_-$ . Given a candidate template set  $V = [v_1, v_2, \dots, v_N]$ , where  $N$  denotes the candidate number,  $v_i$  ( $i = 1, 2, \dots, N$ ) is the  $i$ -th template. With the training template set  $A$ , the sparse representation of  $V$  is formulated as a minimum reconstruction error through a regularized  $\ell_1$  minimization function

$$\min_{\alpha} \|A\alpha - V\|_2^2 + \lambda_1 \|\alpha\|_1 \quad (4)$$

where  $A = [A_+ A_-] \in \mathbb{R}^{K \times (N_p + N_n)}$  is composed of  $N_p$  positive templates  $A_+$  and  $N_n$  negative templates  $A_-$ ,  $K$  is the feature dimension of a template. The parameter  $\lambda_1$  controls the sparsity of the coefficients vector  $\alpha$ .  $\|\cdot\|_2$  denotes the spectral norm.  $\|\cdot\|_1$ , i.e.,  $\ell_1$ -norm, denotes the sum of absolute value of nonzero elements.

##### 3.1.1. Target template

For simplicity, each template has the same size, i.e.,  $8 \times 8$  pixels. At the first frame,  $N_p$  positive templates are sampled around the manually selected target position. Then, these selected templates are normalized to a larger size ( $12 \times 12$  in our experiments) for more details. Each up-sampled image is stacked to form a corresponding target template vector. Similarly, the negative training set is composed of templates further away from the marked posi-

tion. In this way, the negative training set consists of the background.

##### 3.1.2. Feature selection

For a small target template, its gray-scale feature is redundant. As shown in Fig. 1, the training template set  $A$  can also be defined as  $[A_1, A_2, \dots, A_K]^T \in \mathbb{R}^{K \times (N_p + N_n)}$ . Features at the same pixel position of each template are stacked to form a column vector 'sample'. Besides, each element of the vector  $y = [y_1, y_2, \dots, y_{(N_p + N_n)}]^T \in \mathbb{R}^{(N_p + N_n) \times 1}$  represents the property of each template in the training set  $A$ , i.e., +1 for positive templates and -1 for negative templates. In general, the property of each template can be estimated by the linear combination of the template's features, which is represented by

$$y = A^T s + \varepsilon = s_1 A_1 + s_2 A_2 + \dots + s_K A_K + \varepsilon \quad (5)$$

where  $s = [s_1, s_2, \dots, s_K]^T \in \mathbb{R}^{K \times 1}$  is a coefficient vector, and  $\varepsilon$  is the error vector.

To select some discriminative features, the SDC model reconstructs the template property by selecting a relatively small subset from original features.  $s$  can be solved via optimizing a regularized  $\ell_1$  minimization

$$\min_s \|A^T s - y\|_2^2 + \lambda_2 \|s\|_1 \quad (6)$$

which is similar to Eq. (4). Nonzero elements of the solution  $s$  correspond to discriminative features selected from the original  $K$ -dimension feature space.

Then, the original feature space is projected to the selected feature space via a projection matrix  $S$ . It is formed by removing all-zero rows from a diagonal matrix  $S^*$ , where the diagonal elements  $S^*_{ii}$  is zero when  $s_i$  is zero, otherwise, elements are 1.

Both the training template set and the candidate templates are projected to the selected discriminative feature space. Specially, the training template set  $A$  and candidate template set  $V$  in the projected space are  $A^* = SA$  and  $V^* = SV$ .

##### 3.1.3. Confidence measure

Similar to Eq. (1), the projected candidate template set  $V^*$  is represented by the projected training template set  $A^*$  with a coefficients vector  $\beta$ , which is calculated by

$$\min_{\beta} \|A^* \beta - V^*\|_2^2 + \lambda_3 \|\beta\|_1 \quad (7)$$

where  $A^* = [A^*_+ A^*_-]$  is composed of the projected positive template set  $A^*_+$  and the negative templates  $A^*_-$ .  $\lambda_3$  denotes a penalty parameter.

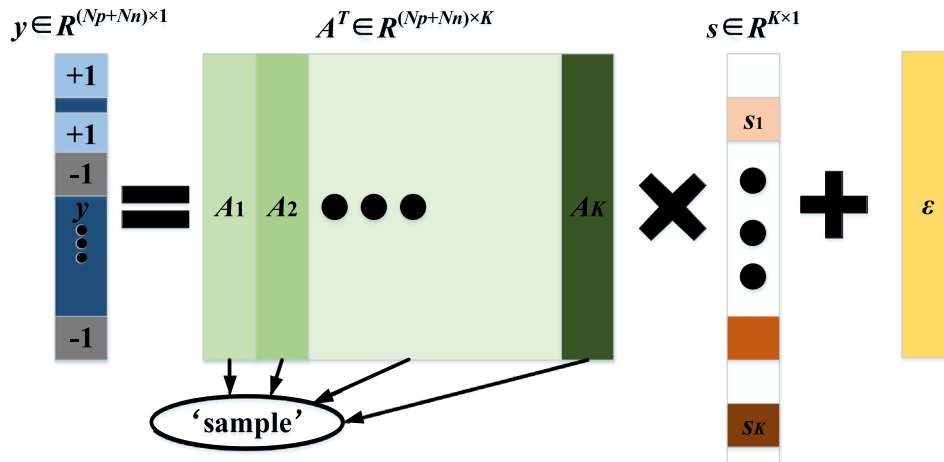


Fig. 1. Feature selection.

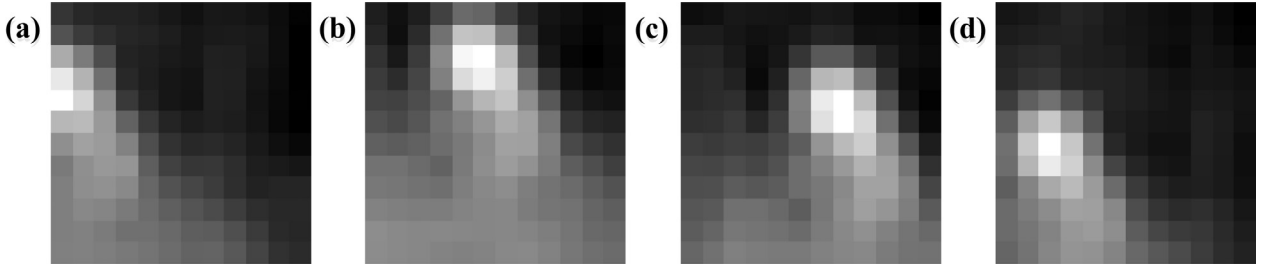


Fig. 2. The original positive templates.

A candidate template with smaller reconstruction error using the positive template set indicates it is more likely to be a target template, and vice versa. Thus, the confidence value  $H_c$  of the candidate template set  $V^*$  can be formulated by

$$H_c = \exp(-(\varepsilon_f - \varepsilon_b)/\sigma_1) \quad (8)$$

where  $\exp(\cdot)$  denotes the exponential function,  $\varepsilon_f = \|V^* - A_+^* \beta_+\|_2^2$  is the reconstruction error of the candidate template set  $V^*$  with the positive template set  $A_+$ , and  $\beta_+$  is the corresponding sparse coefficient vector. Similarly,  $\varepsilon_b = \|V^* - A_-^* \beta_-\|_2^2$  is the reconstruction error of the candidate template set  $V^*$  using the background template set  $A_-$ , and  $\beta_-$  is the related sparse coefficient vector. The variable  $\sigma_1$  is fixed to be a small constant that balances the weight of the discriminative model and the generative model described in Section 3.

The confidence value  $H_c$  gives higher weights to candidate templates that are considered as positive templates (i.e.,  $\varepsilon_f$  is smaller than  $\varepsilon_b$ ) and penalizes others. However, the indistinguishable template (it can be equally constructed by positive and negative template sets) may have a large reconstruction error on target templates. Thus, it introduces ambiguity to the tracker.

### 3.1.4. The prior model

A small target occupies only a few pixels in a target template. However, the target template also contains some background pixels, which have negative influence on the classification performance. In other words, the target template is indistinguishable if it is filled with background clutter or the object is not in the template center.

To solve the problem, a prior model is adopted to obtain another positive template set  $I_j (j = 1, 2, \dots, N_p)$ , which is expressed as

$$I_j = I_{j0} w_{\sigma_2}(u - u^*) \quad (9)$$

where  $I_{j0}$  denotes the image intensity of the  $j$ -th target template in the original set  $A_+$  (see Fig. 2),  $u$  is the position of an arbitrary pixel in a template and  $u^*$  denotes the target position (the template center) at the first frame. The Gaussian weight function  $w$  is described as

$$w_{\sigma_2}(u - u^*) = a \cdot \exp(-(u - u^*)^2 / \sigma_2^2) \quad (10)$$

where  $a$  is a normalization constant,  $\sigma_2$  denotes the distribution of the weight function and is set as the mean of the template's height and weight.

In Eq. (10), the function is motivated by the biological visual system which concentrates on certain image regions. The closer the position  $u$  is to the target position  $u^*$  at the first frame, the more important it is a target pixel, and larger weight should be set. Then, a group of original positive templates are shown in Fig. 2 and the result that they are processed via the prior model is shown in Fig. 3.

As shown in Fig. 2(a), the target in an original sampled target template is away from the template center, which means that the target template is indistinguishable. Easily, the classifier performance can be influenced by inaccurate positive templates. Since

a center pixel has a great possibility to be a target pixel, the prior model is tenable in theory. It is shown in Fig. 3(a) that the target region is blurred, and pixels in the center are enhanced via the prior model. As shown in Fig. 2(b), the target submerges in the background. However, the processed template (see Fig. 3(b)) is more like to be a target template. In short, the prior model aims to obtain more accurate positive templates and is advantageous to reduce the gray intensity of background pixels. Besides, it takes no time for target tracking, for the reason that it is completed before tracking.

### 3.2. The CNGM model

Given a target template in the positive template set  $A_+$ , a hierarchical representation architecture with a convolutional network contains two separated layers. First, local selective features are extracted from a bank of filters convolving the input image at each position. Second, selective features are stacked together to form a target representation.

Each input template is obtained from the target template set and is warped to a canonical size of  $n \times n$  pixels (e.g.,  $12 \times 12$  pixels). The input is represented by the image intensity, which is defined as  $I_j (j = 1, 2, \dots, N_p) \in \mathbb{R}^{n \times n}$ . Here, each input  $I_j$  has been processed via the prior model in Section 3.1.4, in order to obtain a dim target template set. Then, a set of overlapped local image patches  $Y_i (i = 1, 2, \dots, l)$  centered at each pixel position is densely sampled inside the input template through sliding a window of size  $w \times w$ , where  $Y_i \in \mathbb{R}^{w \times w}$  is the  $i$ -th image patch and  $l = (n - w + 1) \times (n - w + 1)$ .

#### 3.2.1. Simple layer

Subsequently,  $N_p$  patches (which are selected in  $N_p \times l$  image patches) are sampled as fixed filters to extract the template features at the first frame. Given the filter  $Y_i \in \mathbb{R}^{w \times w}$ , its response on the input image  $I_j$  is denoted with a feature map  $S_i \in \mathbb{R}^{(n-w+1) \times (n-w+1)}$ , which can be expressed as

$$S_i = Y_i \otimes I_j \quad (11)$$

where  $\otimes$  is the convolution operator.

Furthermore, illustration of obtaining the simple cell feature map is shown in Fig. 4.

As shown in Fig. 4, three templates' simple cell feature maps are obtained. Simple cell feature maps (see the rightmost of Fig. 4) of the target template have a similar geometric layout despite appearance variation, which demonstrates that the local filter  $Y_i$  can extract useful information across the entire image templates. Therefore, the global geometric layout information of the dim target template can also be effectively exploited. For a negative template that contains the cluttered background (see the blue<sup>1</sup> square at the bottom), its feature map undergoes large difference. It is nota-

<sup>1</sup> For interpretation of color in Fig. 4, the reader is referred to the web version of this article.

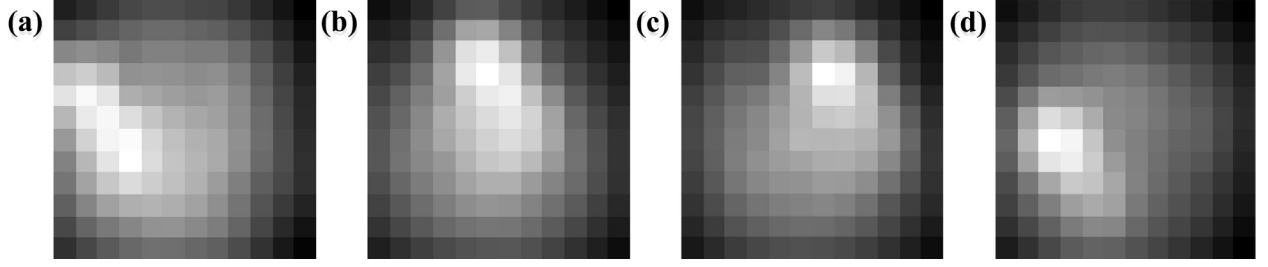


Fig. 3. Positive templates processed via the prior model.

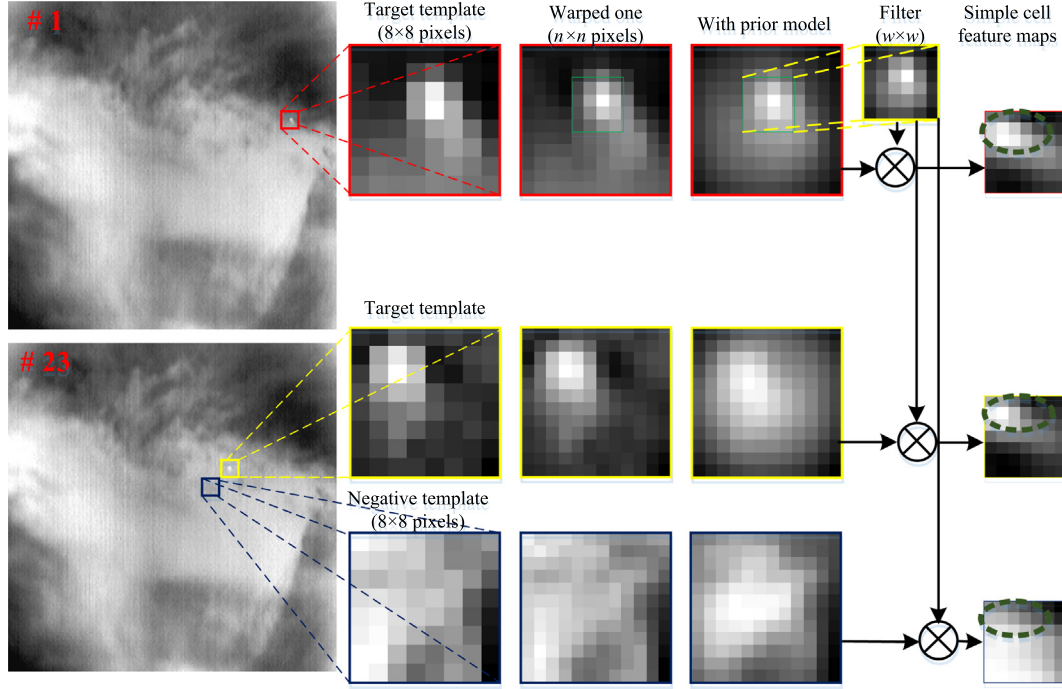


Fig. 4. Simple cell feature maps of three templates.

ble that the prior model helps to obtain a more accurate appearance representation for a dim moving target. In short, the selective filter is localized that can extract local property of a dim moving target.

### 3.2.2. Complex layer

Each simple cell feature map  $S_i$  simultaneously encodes the local structural information of the target template, generating a good representation to handle appearance variations. To enhance the strength of this representation, a three-dimensional tensor  $C \in \mathbb{R}^{(n-w+1) \times (n-w+1) \times Np}$ , which stacks  $N_p$  different simple cell feature maps, is constructed to obtain a complex cell feature map. The illustration of obtaining complex cell feature maps is shown in Fig. 5.

As shown in Fig. 5, the two target templates have a similar complex cell feature map (inside the black rectangle), and the negative template undergoes large difference in the complex cell feature map. This property makes the particle filter (in Section 5) more stable in calculating the weight of particles.

### 3.2.3. Confidence measure

The complex cell feature map can preserve geometric layouts of useful parts as well as local structures. A candidate template with high similarity to the target template in the complex cell feature map indicates that it is more likely to be a target template. Then,

the confidence value  $L_c$  of the candidate template set  $V$  is formulated as

$$L_c = \exp(-\|c_k - c_k^i\|) \quad (12)$$

where  $c_k = \text{vec}(C_k) \in \mathbb{R}^{(n-w+1)(n-w+1)Np \times 1}$  is a column vector by concatenating all the elements in  $C_k$ , which denotes the complex cell feature map of the target at frame  $k$ .  $C_k^i$  represents the complex cell feature map of the  $i$ -th particle at frame  $k$ , and  $c_k^i = \text{vec}(C_k^i)$ .  $\|\cdot\|$  denotes the absolute value operation.

The CNGM model extracts a sufficient image information for a target template, and the complex cell feature map can be considered as the target's feature template. To restrain the ambiguity in the SDC model, a collaborative model utilizing the SDC model and CNGM model is presented. Therefore, the presented tracking algorithm obtains a good appearance representation for a dim moving target.

### 3.3. The SDC-CNGM algorithm for dim moving target tracking

The presented tracking algorithm is formulated within particle filter, which is a Bayesian sequential importance sampling technique and provides a rigorous general framework for dynamic state estimation problems.



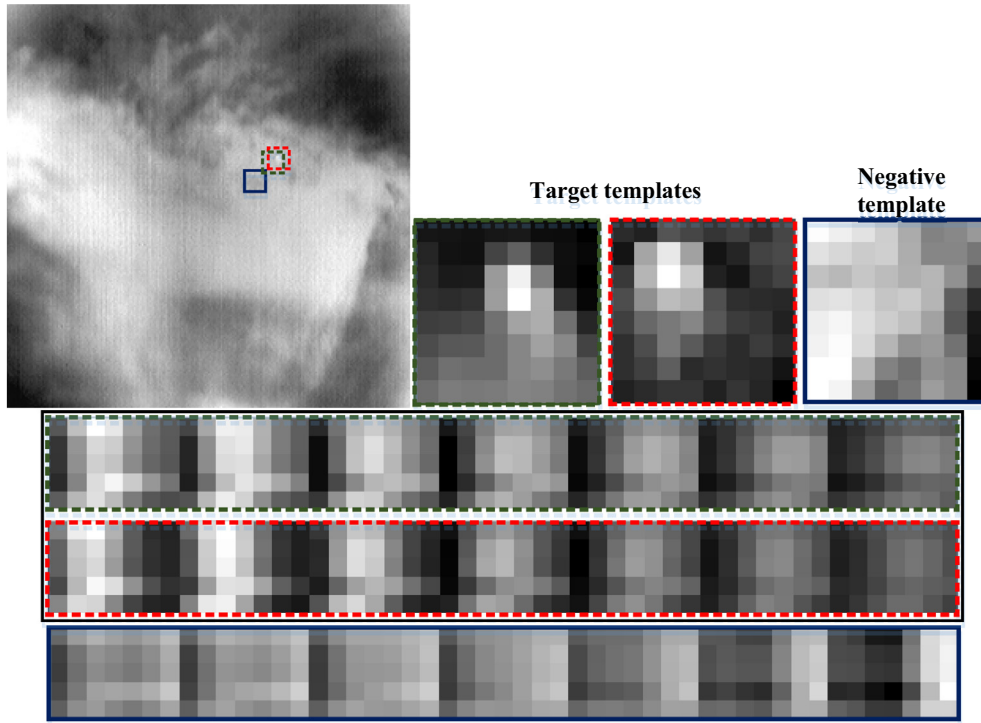


Fig. 5. The complex cell feature of three templates.

### 3.3.1. Transition model and observation model

Since a dim target has little texture information, its sample state vector can be represented by  $x = (x_p, y_p)$ , where  $x_p$  and  $y_p$  denote the coordinate of a particle. To initialize the tracker,  $M$  particles  $x_0(i)$  ( $i = 1, 2, \dots, M$ ) are extracted from the initial PDF  $p(x_0|z_0) \equiv p(x_0)$ , where  $z_0$  means no measurements, and  $p(x_0)$  is a prior that is determined by user input.

In the prediction stage, the target state parameters are independent and can be modeled by two scalar Gaussian distributions, which is formulated as Brownian motion [22]

$$p(x_k|x_{k-1}) = \mathbf{N}(x_k|x_{k-1}, \Sigma) \quad (13)$$

where  $\Sigma = \text{diag}(\sigma_x, \sigma_y)$  is a diagonal covariance matrix, whose elements are the standard deviations of target state parameters.  $\mathbf{N}(\cdot)$  denotes the Gaussian distribution function.

In the update stage, the observation model that estimates the weight for each particle in Eq. (3) is represented by

$$p(z_k|x_k^*(i)) \propto H_c L_c = \exp(-(\varepsilon_f - \varepsilon_b)/\sigma_1 - \|c_k - \hat{c}_k^i\|) \quad (14)$$

where  $H_c$  and  $L_c$  represent particles' responses of the SDC and CNGM model, respectively.

Then, particles' weights can be calculated by Eq. (14). In the presented SDC-CNGM algorithm, these particles work together to find the most possible template in the next frame.

### 3.3.2. Update scheme

Since an object's appearance changes significantly during the tracking process, the update scheme is important and necessary. Tracking with fixed templates is prone to fail in dynamic scenes as it does not consider inevitable appearance change. However, if the template is updated too frequently with new observations, the error is likely to accumulate and the algorithm will drift to the background.

In the SDC model, the positive templates remain the same in the entire sequence. Negative templates are updated every several

frames (5 in our experiments) away from the current tracking result. As the SDC model aims at distinguishing the foreground from the background, it must make sure that the positive templates and the negative templates are all correct and distinct. In this way, the SDC model is discriminative.

For the CNGM model, the positive templates remain the same in the entire sequence and local filters in the first frame are retained. The complex cell feature map  $c_k$  at frame  $k$  is considered as a target template and is updated incrementally to accommodate appearance changes over time, which is expressed as

$$c_{k+1} = (1 - \rho)c_k + \rho\hat{c}_k \quad (15)$$

where  $\rho$  is a learning parameter,  $c_{k+1}$  is the feature vector of a target template at frame  $(k + 1)$  and  $\hat{c}_k$  is the feature vector of the tracked target at frame  $k$ .

The online update scheme not only accounts for appearance variations but also alleviates the migration problem.

### 3.3.3. The SDC-CNGM algorithm

The main steps of the presented algorithm are summarized in Algorithm 1.

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#### Algorithm 1. SDC-CNGM based algorithm

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**Input:** The  $k$ -th infrared image, target template set  $A$ , in SDC model at the first frame.

**Step 1:** At the first frame, the target templates in the set  $A$ , are normalized to the same size, i.e.,  $12 \times 12$  pixels, and are processed via the prior model, i.e., Eq. (6).

**Step 2:** Extracting a filter set from the normalized target template set at the first frame.

**Step 3:** Obtaining complex cell feature maps of target templates by stacking simple cell feature maps.

**Step 4:** At frame  $k$ , a group of negative templates is sampled. Moreover, the projection matrix is calculated via Eq. (3).

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**Algorithm 1** (continued)**Algorithm 1.** SDC-CNGM based algorithm

- Step 5:** At frame  $(k + 1)$ ,  $M$  candidate particles are obtained. Further, complex cell feature maps in the CNGM model and projected candidate templates in the SDC model are calculated.
- Step 6:** Calculating the collaborative confidence value  $H_c \times L_c$  of particles (which have been processed via the prior model) using Eqs. (5) and (9).
- Step 7:** Finding the particle with the maximal weight. Therefore, the center of the particle represents the target position.
- Step 8:** Updating the negative template set in the SDC model and the complex cell feature map of the target template in the CNGM model.
- Output:** Tracking result and updated parameters.

The flow chart of the SDC-CNGM algorithm for infrared dim moving target tracking is shown in Fig. 6.

#### 4. Experimental results and analysis

The performance of the SDC-CNGM algorithm is evaluated by computer simulation. Six challenging infrared image sequences (Seq. 1–6) are used as test sequences, which are collected from diverse sources, e.g., image sequences captured by us, dataset from the Object Tracking and Classification in and Beyond the Visible Spectrum (OTCBVS) benchmark [24], etc. Seq. 1–6 are named *plane 1*, *plane 2*, *plane 3*, *plane 4*, *tank* and *thick cloud*, respectively, which are shown in Fig. 7. The detail of these infrared image sequences and small targets is described in Table 1.

As shown in Fig. 7, Seq. 1–2 and Seq. 6 are obtained by capturing the central  $256 \times 256$  pixels from images taken by a  $640 \times 480$  pixels uncooled IRFPA camera. As shown in Table 1, Seq. 1 consists of 86 frames, a small target submerges in the bright background and moves across the background edge in some images. In Seq. 2, the edge is strong and the target's SCR is lower than 1.5. In Seq. 6, the dim target is added manually, and it undergoes thick cloud. Besides, it moves in a curve and its size is set as  $3 \times 3$  pixels.

Seq. 3 is obtained by capturing the central  $400 \times 450$  pixels from images taken by infrared camera. The sequence contains a target with the maximal speed of about 8 pixels per frame.

Seq. 4 is obtained from the OTCBVS benchmark, a dim target appears in the upper right corner at the beginning and follows a downward diagonal path. Moreover, several false objects appear in the sequence.

In Seq. 5, the target motion is unstable, which is caused by camera shake, and the range of the target size is from  $3 \times 3$  pixels to  $7 \times 7$  pixels.

All Experiments are implemented using MATLAB 2013b on a PC with a Intel(R) Core(TM) i3 3.3 GHz CPU and 4 GB RAM. The presented SDC-CNGM is tested and compared with four algorithms, including the MPF algorithm [11], the improved TMT algorithms [6], the MS-PF algorithm [10], the FCT algorithm [18], the WCF algorithm [1] and the  $\ell_1$ -based algorithm [2].

##### 4.1. Experimental setup

For each sequence, the target position is manually labeled at the first frame. In this paper, the size of the tracking box is  $8 \times 8$  pixels, its center denotes the target position. In the SDC model, to prevent much noise in target templates, the number  $N_p$  is set as a small value 5. The number of negative templates  $N_n$  is set as a suitable value 100. The variable  $\lambda_3$  in Eq. (1) is fixed to be 0.001 and  $\lambda_2$  in Eq. (2) is 0.01. The parameter  $\sigma_1$  is 0.4. In the prior model, the parameter  $\sigma_2$  is fixed to be 8, which is the mean of the target template's height and weight. In the CNGM model, the size of the warped image is set to  $12 \times 12$  ( $n = 12$ ), the filter size  $w$  is set as 6 and the number of the filter is set to be 5 that is equal to  $N_p$ . The learning parameter  $\rho$  in Eq. (10) is 0.95. The standard deviations of the target state in the particle filter are set as follows,  $\sigma_x = 4$  and  $\sigma_y = 4$ , and  $M = 100$  particles as a trade-off between computation efficiency and effectiveness. All parameters are fixed for all experiments.

##### 4.2. Qualitative comparison

The qualitative results of seven tracking algorithms are presented in Figs. 8–13.

In the *plane 1* sequence (see Fig. 8), the MS-PF algorithm undergoes large drift when the dim target moves into the bright cloud completely (e.g., frame #36), which is shown in Fig. 8(b). Besides, the tracking migration can be unrecoverable (e.g., frame #53 and #85). The improved TMT algorithm drifts to the background at frame #36 and #53. The WCF algorithm and the  $\ell_1$  algorithm lose the target at frame #85. The FCT algorithm and the presented algo-

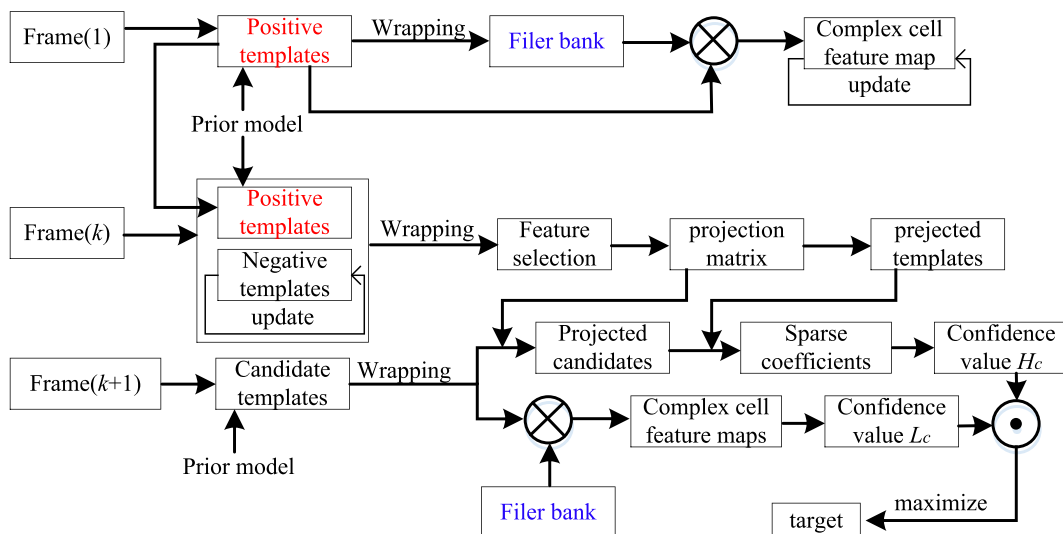


Fig. 6. Flow chart of the presented dim moving tracking algorithm.

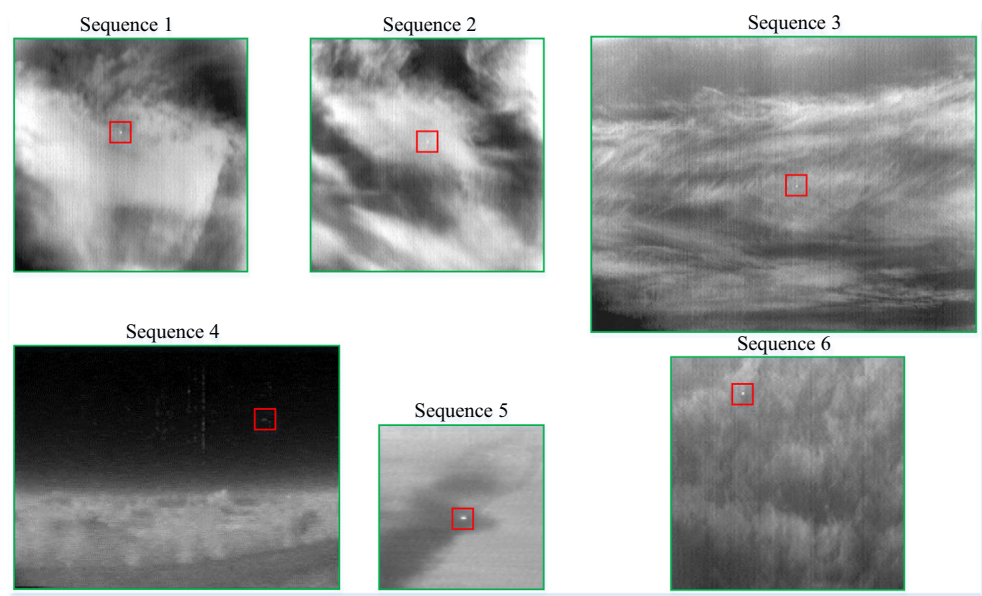


Fig. 7. The six real infrared image sequences.

Table 1  
Overview of test infrared sequences.

| Sequence | Frames | Image size | Target size | Description                                                  |
|----------|--------|------------|-------------|--------------------------------------------------------------|
| Seq. 1   | 86     | 256 × 256  | 2 × 2–4 × 4 | An airplane with bright sky background                       |
| Seq. 2   | 120    | 256 × 256  | 3 × 3–5 × 5 | An airplane with strong cloud edge                           |
| Seq. 3   | 113    | 400 × 450  | 3 × 3–5 × 5 | An airplane with large image size                            |
| Seq. 4   | 200    | 240 × 320  | 3 × 3–5 × 5 | Plane with low gray value, extremely severe background cloud |
| Seq. 5   | 200    | 128 × 128  | 3 × 3–7 × 7 | A tank under the ground background                           |
| Seq. 6   | 200    | 256 × 256  | 3 × 3       | An artificial small target over the thick cloud background   |

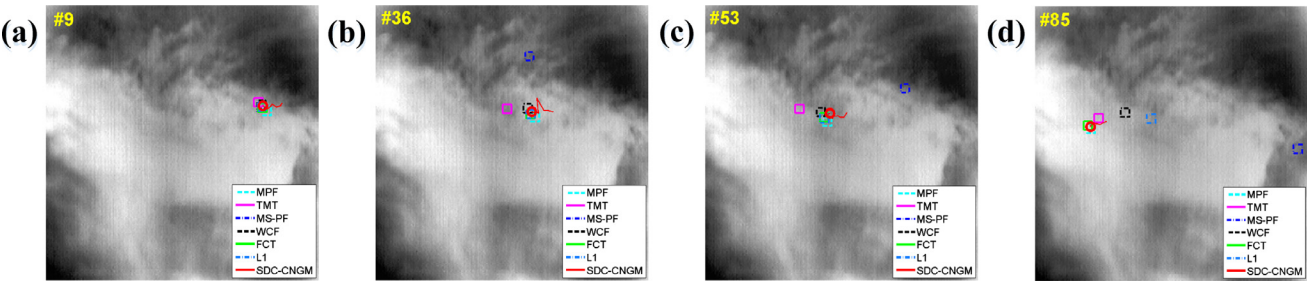


Fig. 8. Qualitative results over plane 1, in which the background is bright. Best viewed on color display. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

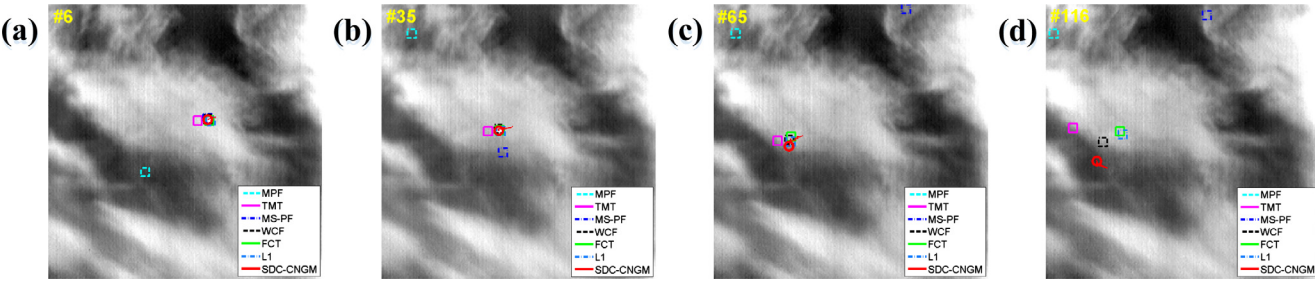
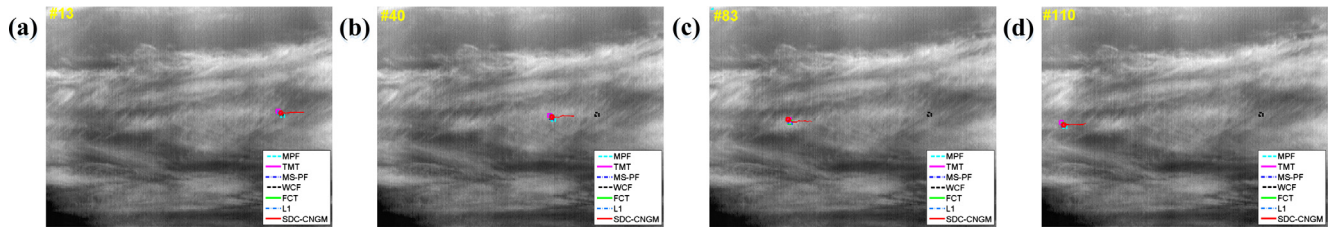


Fig. 9. Qualitative results over plane 2, in which the edge is strong. Best viewed on color display.

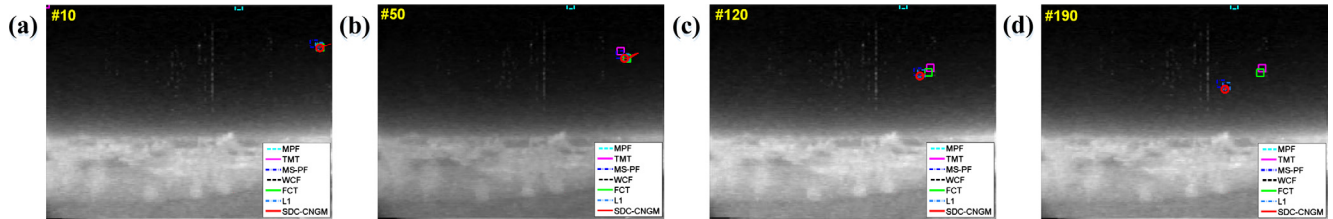
rithm detect the target with a high accuracy. In the plane 2 sequence, the edge of cloud is strong. Only the SDC-CNGM algorithm can track the target well at frame #116. As shown in

Fig. 9, the MPF algorithm is sensitive to the edge of cloud. Moreover, the slow-moving edge easily forms a pipeline, which is mistaken for a target. The TMT algorithm fails to track the target

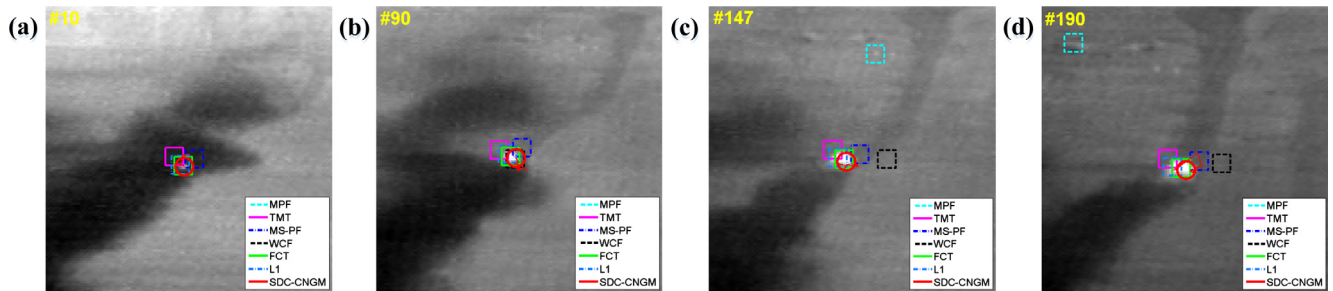




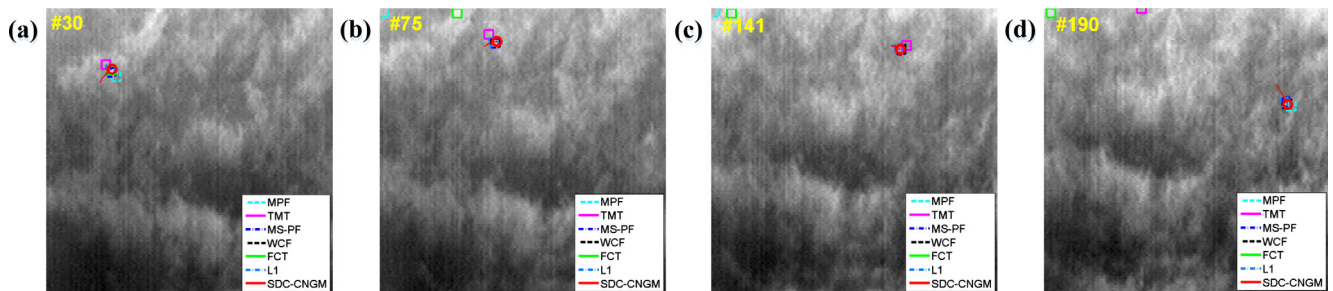
**Fig. 10.** Qualitative results over *plane 3* with large image size. Best viewed on color display. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



**Fig. 11.** Qualitative results over *plane 4* with low gray value. Best viewed on color display. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



**Fig. 12.** Qualitative results over *tank* with ground background. Best viewed on color display. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



**Fig. 13.** Qualitative results over *thick cloud*. Best viewed on color display. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

when it moves cross the strong edge (e.g., frame #65) and the problem is serious at frame #116. The FCT algorithm, the WCF algorithm and the  $\ell_1$  algorithm have a similar tracking results to the TMT algorithm. In the *plane 3* sequence, the image size is large and the target moves fast. The MPF algorithm fails to detect the target at frame #83 (see Fig. 10(c)) and the WCF algorithm loses the target at frame #40, #83 and #110. In the *plane 4* sequence, the background has a low gray value and the space target disappears behind the horizon. Tracking result of the MPF is unacceptable. The TMT algorithm and FCT lose the dim target after frame

#120, and a false target is mistaken for a true target at frame #190. In the *tank* sequence, the camera has a sever shake. The MPF algorithm has frequent detection error (e.g., #147 and #190), due to continuous motion of background pixels. The TMT tracker may drift to background, however, it is possible to detected the target again when the target goes into the target box. Benefiting from the particle filter, the MS-PF and the SDC-CNGM algorithm can obtain a target template with maximal weight from the particle set. The FCT algorithm can track the target well, due to the fact that the feature of a target with a larger size can be dis-

criminative and the ground background contains less clutter. Besides, the WCF algorithm drifts to the background at frame #147 and #190. Tracking results of the FCT algorithm and SDC-CNGM algorithm over Seq. 5 are satisfactory (see Fig. 12). In the *thick cloud* sequence, the target submerges in the cluttered background in some images. The MPF tracker tends to detect the edge of the moving cloud as a target (e.g., frame #75 and #141). The

TMT also drifts to the background at frame #190, due to that the dim target template has a little intensity information. The FCT algorithm loses the target and the accumulative error becomes larger and larger, which is a bottleneck of most tracking algorithms without using detection.

The SDC-CNGM tracker has done well in tracking a dim moving target regardless of the bright background, the ground background

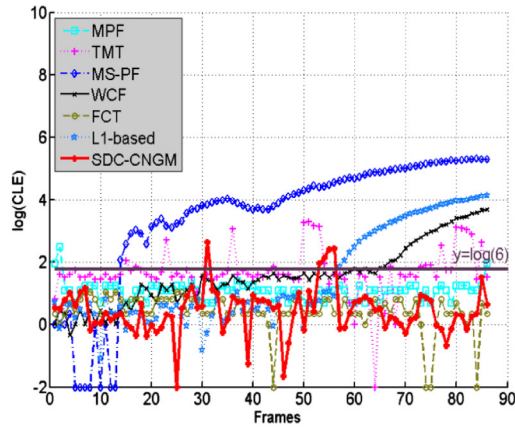
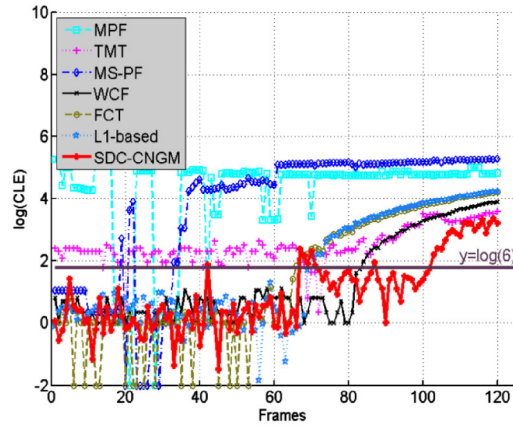
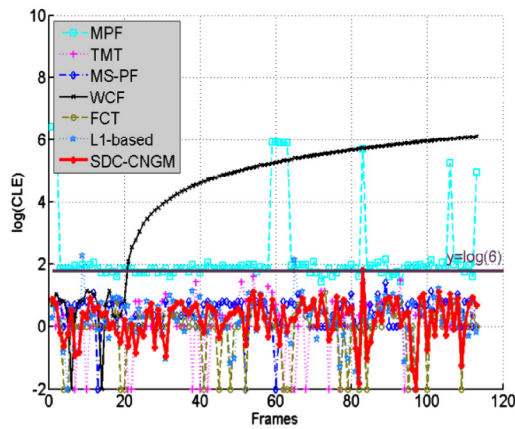
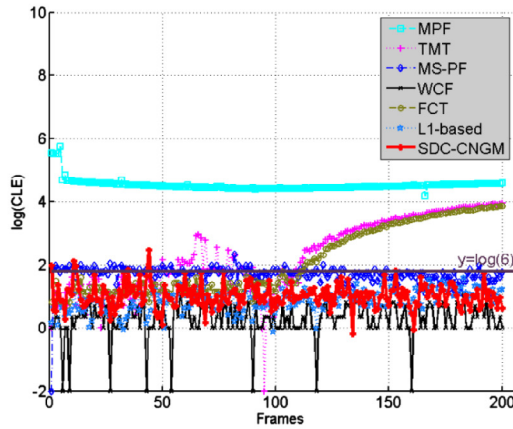
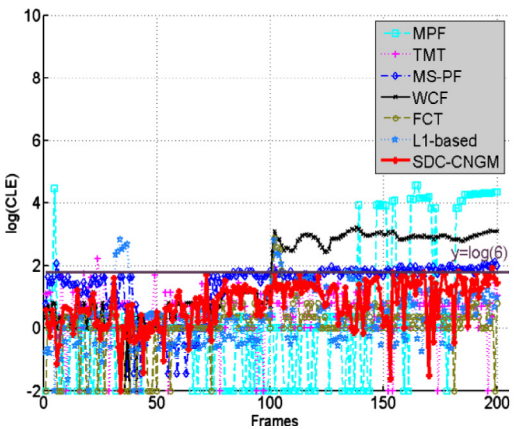
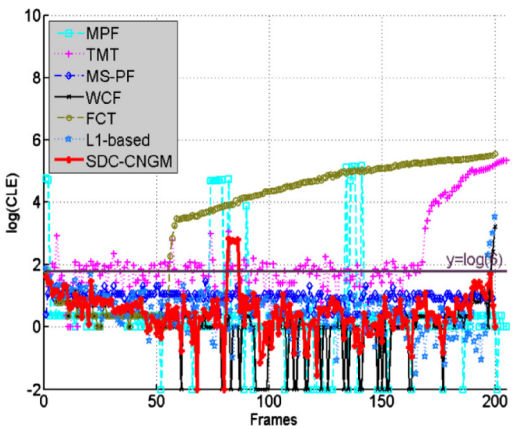
(a) *plane1*(b) *plane2*(c) *plane3*(d) *plane4*(e) *tank*(f) *thick cloud*

Fig. 14. The logarithm value of the CLE.

or space background, for the reason that enough target information is extracted, and the distinction between the target and the background is utilized.

#### 4.3. Quantitative comparison

For further quantitative analysis, a similar standard evaluation in [25] is adopted, containing Center Location Error (CLE), Success Rate (SR) and Frame Per Second (FPS). The CLE denotes the Euclidean distance  $d$  between the estimated target and the true target. The SR is defined as  $n_1/s_1$ , where  $s_1$  is the frame number of a sequence and  $n_1$  is the frame number on condition that  $d < 6$ . The FPS denotes the number of the image which is processed in one second on a PC. Therefore, the CLE and the SR represent the accuracy while the FPS represents the algorithm speed. Then, a group of experimental results is shown in Fig. 14 and Table 2.

In Fig. 14, the vertical axis represents the logarithm of the CLE value, for a suitable scale.  $\log$  represents the logarithm operator and the line ' $y = \log(6)$ ' is a boundary (which is equal to that the CLE value is 6). As shown in Fig. 14(d), the CLE curves of the MPF algorithm over the *plane 4* sequences are relatively unacceptable. In Fig. 14(a), The MS-PF tracker has a large fluctuation after frame #15, which is because the target moves into the bright cloud. Besides, the  $\ell_1$  algorithm and WCF algorithm drift to the background after #65 and the excursion can't be recoverable. In Fig. 14(b), only the SDC-CNGM algorithm obtains a satisfactory result at most frames and other algorithms lose the target in dozens of frames, for the reason that the *plane 2* sequence contains strong edge and a target with a low SCR. With severe background clutter, the target template in the TMT algorithm can't be updated ideally, leading to the cumulative error (see Fig. 14(b) and (d)). The MPF algorithm regards the moving edge as a target easily. Therefore, its false alarm rate increases. It is indicated in Fig. 14(d) that the TMT algorithm always loses the target and the PF-based algorithms (the MS-PF and SDC-CNGM algorithms) can deal with the cluttered background well. Since the target in Seq. 4 has low gray value, the noise point is easily detected as a target for the MPF algorithm. Moreover, the TMT algorithm and the FCT algorithm are both influenced by the false target and the corresponding migration is unrecoverable, which is shown in Fig. 14(d). In Fig. 14(e), camera shake makes the target unstable. Nevertheless, the CLE results of the algorithms (except the WCF algorithm) are satisfactory, due to that the target information is sufficient. The WCF algorithm is adaptive to kinds of backgrounds, however, it can't adapt to the situation that the target moves fast and it is out the scope of the current search region (see Fig. 14(c) and (e)). In Fig. 14(f), since the target template is influenced by the thick cloud, the CLE curve of the TMT algorithm is unstable. The MPF algorithm has tracking migration in several frames, which is because the background moves slowly. The FCT algorithm loses the target at frame #50 and drift to the far position in a short time. The CLE values of the SDC-CNGM tracker over all infrared

sequences are all small, demonstrating that the presented algorithm can track the dim moving target well and be adaptive to kinds of infrared cluttered backgrounds.

As shown in Table 2, the success rate result of the MPF over *plane 4* sequence are 0, due to that this algorithm is sensitive to the environment. For the TMT algorithm, migration in the matching process results in an unrecoverable detection (see Fig. 14 (b) and (d)) and its success rate over the *plane 2* sequence is 0.08. The MPF algorithm has a modest success rate (up to 96%), however, it is time-consuming. The FPS is 5 and the value can be smaller when the image size is larger. Besides, it is greatly influenced by the edge (see Fig. 14(b)) and the success rate over Seq. 2 decreases to 0.12. Since the MS-PF algorithm utilizes the gradient histogram feature of an infrared image template, it can't obtain a stable gray statistics when the target submerges into the bright cloud (e.g., Seq. 1 and Seq. 2). For example, in the *plane 1* sequence, its success rate are 14%, which are much smaller than that in Seq. 3–6. Since the WCF algorithm searches the target in a local region, a fast target's position may not be estimated accurately. For example, in the Seq. 3 and Seq. 5, the success rate values are 18% and 50%, which are much smaller than that in the other sequence. The FCT has a low success rate when the target submerges in the cluttered background (Seq. 2 or Seq. 6), resulting in an inaccurate classification. Moreover, the  $\ell_1$  algorithm utilizes the appearance model in the FCT algorithm with the  $\ell_1$  constraint. In Table 2, its average SR value is greater than that of the FCT algorithm.

The SDC-CNGM algorithm does well in tracking a dim moving target over kinds of complex backgrounds. On the one hand, the SDC model takes the background into consideration, which makes the algorithm be discriminative. On the other hand, the CNGM model obtains sufficient target information even if the target has a low gray value (e.g., *plane 2* or *plane 4*) or undergoes a severe background (e.g., *plane 4* or *thick cloud*). Besides, the influence of the clutter background in a target region can be effectively restrained by the prior model.

Experimental results insist that the presented algorithm is real-time on a PC (the average FPS is 28.5), due to the fact that the SDC model extracts several templates and the CNGM model deals with convolution operation in the Fourier domain. In term of subjective vision and objective evaluation, the SDC-CNGM algorithm is superior to the other six algorithms in infrared dim moving target tracking.

#### 4.4. Comparison of the SDC model and the CNGM model

While the SDC model is utilized to extract a discriminative representation of a dim moving target, the CNGM model that considers the local structural information of a dim moving target is adopted for object representation. In theory, the two models are complementary. Therefore, a collaborative model that integrates them is presented for dim-small target tracking. Moreover, another group of comparison results is shown in Fig. 15.

**Table 2**

Success Rate (SR) (%) and average frame per second (FPS). **Bold** fonts indicate the best performance while the *italic* fonts indicate the second best ones. The total number of evaluated frames is 919.

| Sequence           | MPF  | TMT  | MS-PF    | WCF         | FCT       | $\ell_1$ | SDC-CNGM    |
|--------------------|------|------|----------|-------------|-----------|----------|-------------|
| <i>Plane 1</i>     | 0.96 | 0.77 | 0.14     | 0.72        | <b>1</b>  | 0.65     | 0.96        |
| <i>Plane 2</i>     | 0.12 | 0.08 | 0.24     | 0.69        | 0.54      | 0.58     | <b>0.8</b>  |
| <i>Plane 3</i>     | 0.65 | 0.98 | <b>1</b> | 0.18        | <b>1</b>  | 0.98     | <b>1</b>    |
| <i>Plane 4</i>     | 0    | 0.47 | 0.91     | <b>1</b>    | 0.56      | <b>1</b> | 0.98        |
| <i>Tank</i>        | 0.81 | 0.99 | 0.91     | 0.5         | 0.98      | 0.94     | <b>1</b>    |
| <i>Thick cloud</i> | 0.92 | 0.59 | 0.93     | <b>0.99</b> | 0.27      | 0.97     | 0.97        |
| Average SR         | 0.58 | 0.65 | 0.69     | 0.68        | 0.73      | 0.85     | <b>0.95</b> |
| Average FPS        | 5    | 23   | 21       | 34          | <b>40</b> | 13.5     | 28.5        |



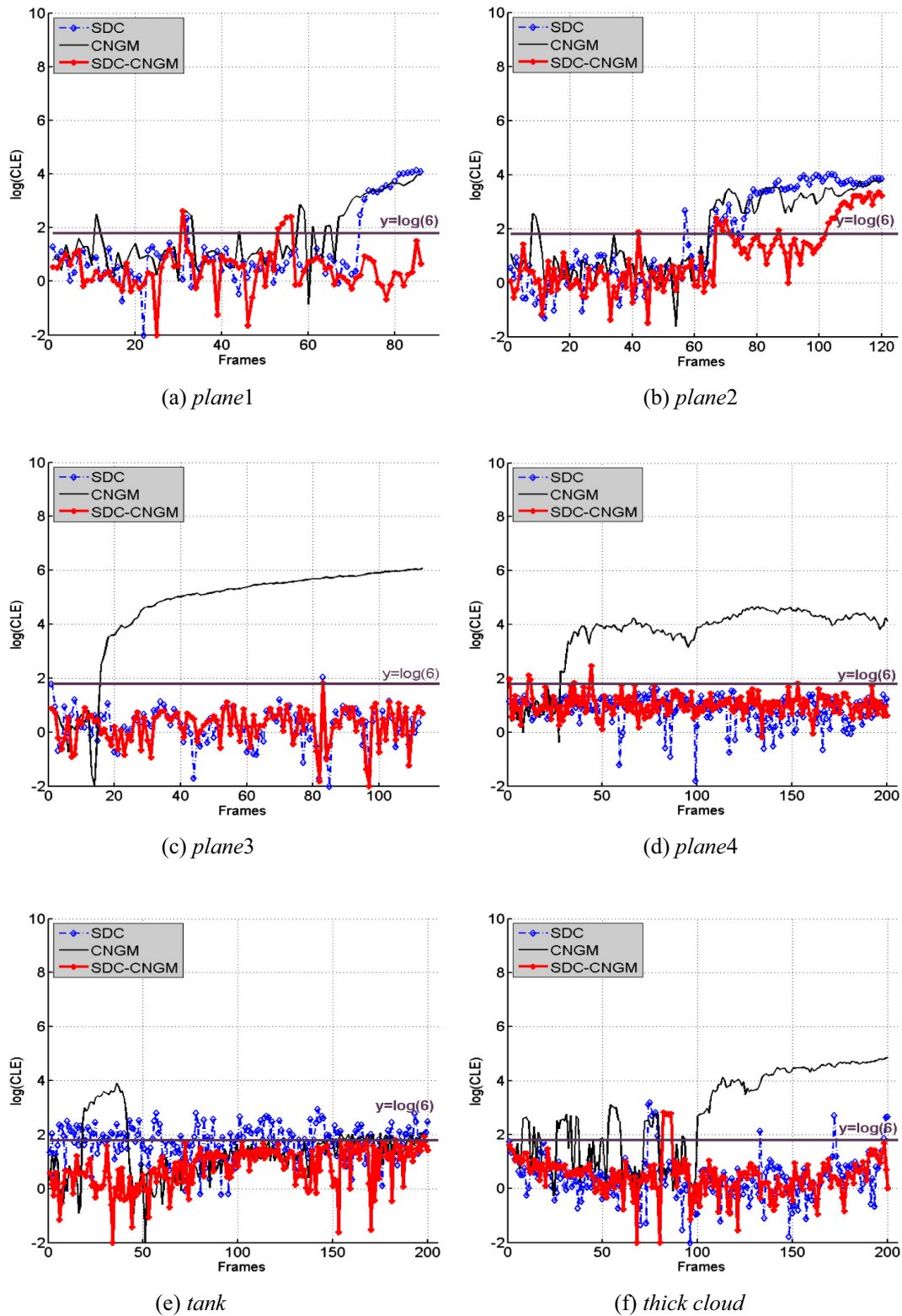


Fig. 15. The logarithm value of the CLE.

As shown in Fig. 15, the CLE curve of the CNGM model has a larger fluctuation than the SDC model or the collaborative model. Specially, the generative model is sensitive to the background, due to the fact that the dim moving target has weak features. Besides, the SDC model distinguishes the target from the background. Therefore, the SDC model based tracking algorithm loses

the target while the small target undergoes strong background and edge, which is shown in Fig. 15(a) and (b). In Fig. 15(e), the CLE result of SDC model indicates that the target and the background cannot be well distinguished as the target motion is unstable caused by camera shake. However, the CNGM model obtains an ideal result, which is because the target information is sufficient.



Then, the proposed SDC-CNGM algorithm gives accurate tracking results. In short, the two models are complementary to each other and the performance of the collaborative model is better than the two models at most time.

## 5. Conclusions

In this paper, an algorithm based on sparsity-based discriminative classifier and convolutional network is presented for infrared dim moving target tracking. In the SDC model, holistic templates are incorporated to construct a discriminative classifier that can effectively deal with the cluttered background. In the CNGM model, the efficient convolutional network generates an strong representation, which encodes the local structural information of a small target template. Besides, the prior model helps to restrain the influence of the background clutter. The contribution of these discriminative and generative models are integrated in a unified manner. Moreover, the online update scheme makes the presented algorithm adapt to kinds of complex backgrounds. At last, experimental results demonstrate that the presented algorithm outperforms other state of the art algorithms in the accuracy, and, broadly, makes great contributions on laying the foundation for future recognition.

## Acknowledgements

We would like to express our sincere appreciation to the anonymous reviewers for their insightful and valuable comments, which have greatly helped us in improving the quality of the paper. This work is partially supported by the National Natural Science Foundation of China (No. 61675160, 61265006 and 61401343), the 863 Program of China (2014AA8098089C), and Chinese Academy of Sciences (LSIT201503).

## Conflict of interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service or company that could be construed as influencing the review of the manuscript entitled '*Infrared Dim Moving Target Tracking via Sparsity-based Discriminative Classifier and Convolutional Network*'.

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